# Review status after V9.0 · the layout freeze

Taif / Palm House · the design review of 30 Oct 2026 (`docs/review/design_review.typ` → `output/Design_Review_V6.pdf`, tag `review-1`, written against V6.2) and the interim verification and validation review of V8.3 (3 Nov 2026, `archive/VV_Review_V8.3.pdf`) · status on 6 Nov 2026 after **V9.0, the layout freeze**: the site set out on site (day 1, the diagonals exact) and the final position (1.24 + 0.50 m south, B1 5.49 m from M1); the cut wall 0.12 m shorter (field 2.30 m; 0.42 m from the wall as built, 0.90 m from the deed); the whole-farm base v2 with the old house, the covered parking, the tanks and the septic placed (approximate); bedroom 1 for young children without an en-suite · the rows updated in V9.0: S-03, G-04, E-03, S-01, S-02, S-08 and V-01, V-03, V-06, V-07 · every critical and major finding: 40 in all, **resolved** 21 · **partly** 12 · **open** 7

## 1 · The design review (V6): 30 critical and major findings

| Id | Sev. | Where | Status | Version | What was done · what is still open |
|---|---|---|---|---|---|
| **Y-01** | critical | Courtyard · the envelope | **resolved** | V7.0 / V7.1 | Enclosure option A: W1 on the gallery edge, W2 on the family living's edge, W3 over the living's courtyard glass, black steel divided lites, the projection 1.55 m (B3 / R11 / C21–C23 at x 5.80); the courtyard open to the sky; semicircular heads over the three-arch arcade. |
| **A-01** | critical | Ageing · the lift | **resolved** | V6.3 | The lift installed at construction, two stops; the shaft a placeholder until a supplier's drawing (open item: headroom ≤ 3.40 m / ≤ 3.70 m with a raised head). |
| **S-03** | critical | Site · north and lane setbacks | **partly** | V9.0 | **Set out on site** by the owner (day 1, diagonals exact) and placed: 1.24 m + 0.50 m further south (B1 5.49 m from M1) and the cut wall 0.12 m further out after the field reading of 2.30 m at the tightest point: the margin **0.90 m from the deed setback line, 0.42 m from a setback counted from the wall as built** (the wall 0.62 m inside the deed line). **Open:** the official survey and the engineering office's answer on which line the setback is counted from. |
| **E-08** | major | Services · fire and life safety | **resolved** | V8.0 / V8.1 | Home-level strategy (owner): interconnected mains smoke alarms with battery back-up, heat, CO and gas detectors, extinguishers and a blanket; escape casements without screens in every sleeping room; thumb-turns on the exits; the first floor's exits the main stair plus the escape windows; the suite court a fresh-air area, not an exit. |
| **G-01** | major | GF · living wall (west) | **resolved** | V6.3 | The solid TV wall with the entrance arch and the arched TV niche. |
| **G-04** | major | GF · maid room | **resolved** | V7.0 / V9.0 | The kitchen's north wall at y 7.0 (V7.0); V9.0: the cut wall 0.12 m further out: the maid room 7.69 m² clear (was 7.28), the maid bath 3.24 m². |
| **F-01** | major | FF · gallery width | **resolved** | V7.0 | The projection 1.55 m: the gallery 1.405 m clear, 1.235 m at the column sleeves. |
| **F-02** | major | FF · door widths | **resolved** | V6.3 | Bedroom and bathroom doors 1.00 m openings; exceptions recorded. |
| **F-06** | major | FF · family living | **resolved** | V7.0 | W2 on the courtyard edge: a conditioned room. |
| **F-07** | major | FF / roof · roof access | **resolved** | V7.0 / V8.1 | A manufactured maintenance ladder in a recess of a stone pilaster on the gym's east wall in the suite court, behind a locked walnut panel; the guarding continuous to every piece of equipment. **Open:** the fall protection confirmed by the specialist. |
| **X-01** | major | Circulation · serving the majlis | **resolved** | V8.1 (owner) | Accepted by the owner: the food route kitchen → dining → riwaq → serving passage stays; very large events use the old majlis (inside the old house). |
| **G-02** | major | GF · living + dining | **partly** | V7.0 / V8.2 | The courtyard well decided; the clerestories get the winter noon sun and stay openable; no garden door (owner). **Open:** the daylight simulation. |
| **G-03** | major | GF · kitchen | **partly** | V8.0 | The hood's make-up air: a filtered motorised inlet interlocked with the hood. **Open:** the MEP sizing. |
| **G-07** | major | GF · bathrooms | **resolved** | V8.3 (owner) | No ground-floor shower (owner): ageing in place relies on the lift and the first-floor bathrooms; the family powder room (two stone basins, a wall-hung WC). |
| **Y-02** | major | Courtyard · size for daily use | **resolved** | V8.0 / V8.3 | The level courtyard with movable seating for 4–6 facing the TV, the water + tree enclosure in front of the kitchen window, the north side free. |
| **Y-03** | major | Courtyard · sunken seating safety | **resolved** | V8.3 | No step and no pit: the whole floor level with falls to linear drains and an overflow. |
| **T-01** | major | Structure · X5 | **partly** | V8.0 | Load path revised, **improved, not verified** (R6 removed; B19, B22, the office wall on rib B21). **Open:** the engineer. |
| **T-03** | major | Structure · soil and foundations | **open** | | No soil report: the pads, the lift pit, the breeze channel and the soakaways are assumed. |
| **E-01** | major | Services · the malqaf's value | **resolved** | V8.0 | Air system v3 / v3.1: one tower with a courtyard-breeze path and the HRV intake in its base on the roof. The MEP engineer sizes it. |
| **E-02** | major | Services · ventilation and dust | **partly** | V8.0 | A 750 m³/h class HRV with an ePM1 ≥ 50 % filter, continuous + boost, a dust mode (assumed). **Open:** the MEP design. |
| **E-03** | major | Services · plumbing | **partly** | V8.3 / V9.0 | The owner's decisions (an external east stack, trap primers, rain / sewage separate, the condensate by tundishes); bedroom 2's en-suite and the family WC on one service wall; the powder room; V9.0: the septic serves both houses, a chamber at every bend, the inlet −1.24 on the computed falls. **Open:** the suite bath's drainage approach, the water system (tankers, tanks, roof tank, pumps, greywater), the plumbing engineer. |
| **E-04** | major | Services · solar PV provision | **resolved** | V8.1 / V8.2 | PV zones inside the guarded roof zones; the conduits down the west shaft to the inverter and batteries in the utility room (the electrical engineer confirms). |
| **E-05** | major | Services · generator and essentials circuit | **resolved** | V8.2 (owner) | No generator: the backup from the PV and the batteries when installed; an essentials circuit at the main panel. |
| **E-06** | major | Services · security and the safe room | **resolved** | V8.1 / V8.2 | No safe room (owner); one site entry, the farm's 4 m main gate; an intercom and seven cameras, the alarm panel in the utility room. |
| **E-07** | major | Services · acoustics of the living | **open** | | The double-height living's finishes and absorption at the details round. |
| **S-01** | major | Site · parking | **partly** | V9.0 (owner) | The family's **existing covered parking for two cars is south of the old house, west of the driveway** (owner, 5 Nov 2026; approximate, to be measured); one uncovered family car in the north strip, kept (now 2.04 m off the north wall). **Open:** where guests park (the brief's farm parking south of the build land is withdrawn). |
| **S-02** | major | Site · guest approach | **resolved** | V8.2 / V9.0 | The guest path: a 1.2 m honey-stone band along the driveway's east edge from its south end to the guest entrance (13.75 m, bollard lights), never entering the family areas; the old majlis guests go along the driveway to the old house. |
| **S-04** | major | Site · garden access | **resolved** | V8.2 (owner) | No garden door from the living / dining (owner): the garden is reached round the house. |
| **S-05** | major | Site · east strip, service yard | **resolved** | V8.3 | The outdoor counter next to the service door, the grill and tannour apart, the screen, the gas cabinet ≥ 3 m from the grill, the bins; the workers along the farm's boundary wall. |

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| **S-08** | major | Site · tanks, septic, plant | **partly** | V9.0 (owner) | The owner's answers placed: the **existing ground tank south of the old house, west of the driveway near peg R**, the reserved tank beside it, the pipe across the driveway to the utility room (27.8 m); the **septic outside the north wall near C1 serves the old house and takes the new house's drains** (the main 38.4 m, the inlet −1.24). **Open:** measure them on site (the tank's lid and fill point, the septic's position, inlet level, capacity and condition), the levels, the municipality's consent. |

## 2 · The interim V&V review (V8.3): 10 critical and major findings

| Id | Sev. | Where | Status | Version | What was done · what is still open |
|---|---|---|---|---|---|
| **V-01** | critical | Site · the setback margin | **partly** | V9.0 | As S-03: set out on site, the final position and the shortened cut wall give 0.90 m from the deed setback line and 0.42 m from the wall as built (field 2.30 m, a 2 cm tape cushion over the owner's 0.40 m target). **Open:** the official survey; the office's answer on the setback line. |
| **V-02** | critical | Structure · unverified, no soil report | **open** | | Unchanged: every structural size assumed (X5, X21, the projection, the ribbed slabs), no soil report. **Next:** the engineers (item 6 below). |
| **V-03** | major | FF · bedroom 1 without a shower | **resolved** | V9.0 (owner) | **Bedroom 1 (next to the master suite) is the room for very young children**; they are bathed in the master suite's bathroom; no en-suite (option A kept; `studies/bedroom1_bath.typ` B / B4 and `studies/bedroom1_ensuite.typ` not adopted). The family WC upstairs stays WC + basin. The kids' room keeps the gallery's WC; a shower for it is revisited only if it becomes a bedroom. |
| **V-04** | major | Services · no heating, no energy-code envelope | **open** | | Unchanged: the heating strategy and the envelope specification (insulation, glazing U-values, the steel windows' thermal breaks) for the MEP engineer and the facades round. |
| **V-05** | major | Services · water and power in a power cut | **open** | | Unchanged: no roof tank in the water scheme yet. **Next:** the water system study (item 3 below: tankers, the ground tanks, a roof tank, pumps, greywater). |
| **V-06** | major | Site · the septic and the long foul main | **partly** | V9.0 (owner) | The owner confirmed the septic near C1 serves the old house and takes the new house's drains; the route recomputed with a chamber at every change of direction, the driveway crossed once at the gate, the inlet −1.24 (≈ 0.84 m below the assumed ground at the gate). **Open:** the surveyed inlet level, the capacity for both houses, a pump if needed, the legal consent. |
| **V-07** | major | Site · the north parking bay against the north facade | **partly** | V9.0 | The house moved 1.74 m south since V8.3: the bay now 2.04 m off the north wall (was 0.4), the swept path clears the wall by 2.39 m. **Open:** the bay still sits in front of the women's majlis north window; the facades round (R-37) decides its exact place or a planted edge. |
| **V-08** | major | East yard · the hood exhaust beside the outdoor counter; the bins at the maid's window | **open** | | Not changed in V9.0: the hood outlet and the make-up inlet beside the counter's south end, the bins ≈ 1 m from the maid's escape window. **Next:** the MEP engineer (the outlet's position and height) and the facades round. |
| **V-09** | major | Fire · one open stair in the riwaq | **partly** | V8.0 (owner) | The owner's accepted residual risk (home-level strategy: early detection, interconnected alarms, escape windows). **Open:** the review's mitigations (a door between the kitchen and the dining, the clerestories' opener) at the details round. |
| **V-10** | major | Suite bath over the kitchen · drainage approach; the utility room | **open** | | Undecided (a: every drain in the depressed slab / b: the east service wall); the utility room still flagged small. **Next:** item 4 below. |

## 3 · The next items, in this order

1. **The official survey and the engineering office's answer on the setback** (from the deed line or from the wall as built): S-03, V-01.
2. **Measure the existing farm on site:** the old house and its annex (W1–W4, A1–A4), the covered parking (P1–P4), the tanks (K0–K2, the lid and the fill point) and the septic's inlet level (`docs/SITE_SETOUT.md` § 8): S-01, S-08, V-06.
3. **The water system study:** the tankers, the ground tanks, a roof tank, the pumps, the greywater lines to the south garden with diversion chambers (V-05, E-03).
4. **The suite bathroom's drainage approach** (a / b) and the utility room (V-10, E-03).
5. **The full verification and validation review of V9.0.**
6. **The engineers:** structure X5 (T-01, V-02), the soil report (T-03), the MEP sizing (G-03, E-01, E-02, V-04, V-08), the lift supplier.
7. **The bathrooms' interior study.**
8. **The facades and the details, acoustics included** (R-37, E-07, G-02, V-07, V-09).

Minor findings, suggestions and strengths are unchanged in both reviews; of the minors, T-02 (X3) is resolved (V6.3), F-09 (the treadmill) in V8.0, Y-04 (courtyard dust and water) answered by the drainage section, **F-05 (bedroom 1's bath) is resolved in V9.0 by the owner's decision** (the young children's room, bathed in the suite). This page is a status update, not a new review. Nothing here is for construction: concept design, all sizes assumed until the engineers confirm.